

FMCG ERP — Administration Manual v2

Version: 2.1 (post-ERP-updates) **Date:** 2026-06-05

Audience: System Administrators, IT Managers, ERP Implementation Teams

Modules Covered: Admin · Users · Roles · Permissions · Security · System Configuration · Companies · Approval Chains · Integrations · AI

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1. Module Overview

What it does: Full system administration — user accounts, roles, permissions, security policies, system configuration, company setup, approval workflows, and external integrations.

Who uses it:

- System Administrator — manages all users, roles, and system settings
- IT Manager — configures integrations, security, and system parameters
- Implementation Team — initial setup of company, chart of accounts, master data

When to use it:

- When onboarding a new user
- When configuring a new department's approval chain

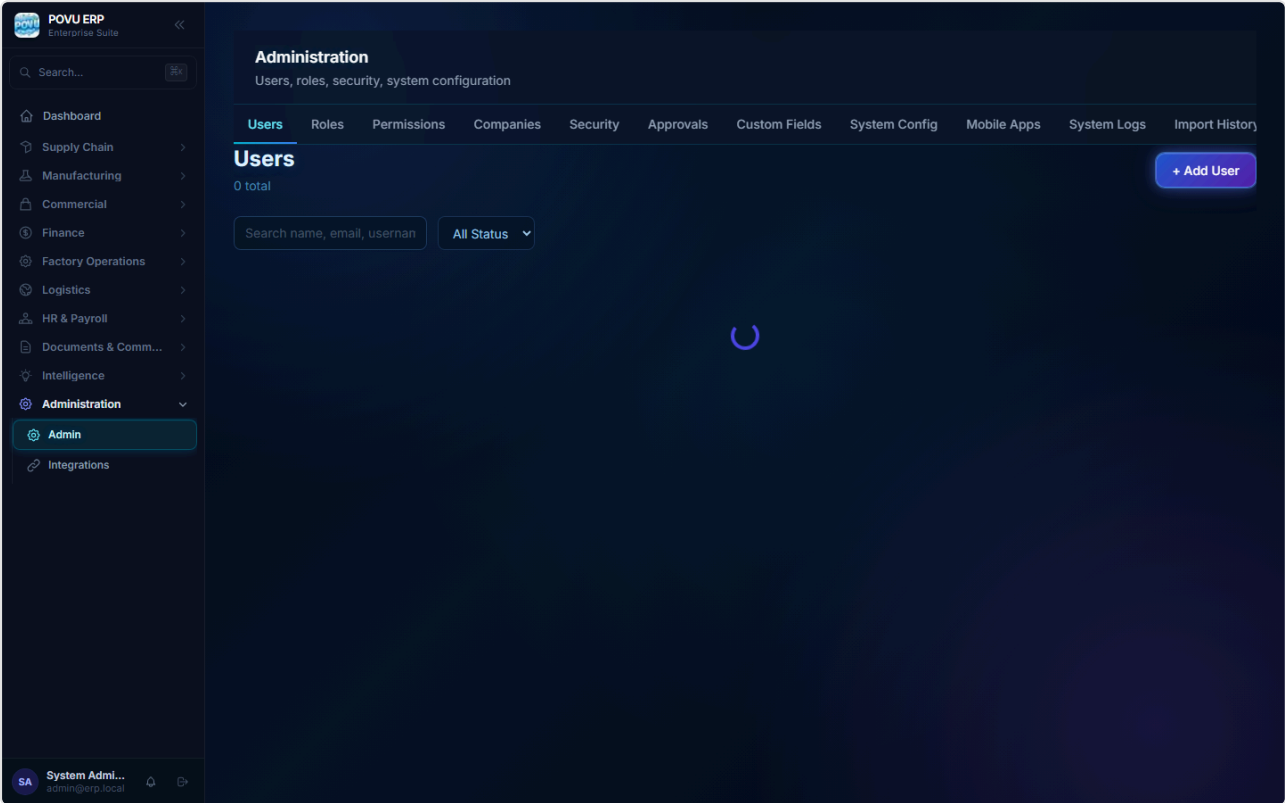
- When troubleshooting a permission error
- When setting up an integration (eTIMS, M-Pesa, bank)
- When resetting a user password

Modules at a glance:

Feature	Route	Purpose
Admin	/dashboard/admin	Full admin workspace
Products	/dashboard/products	Product master
Materials	/dashboard/materials	Materials master
Integrations	/dashboard/integrations	External system connections
AI	/dashboard/ai	AI assistant and configuration
Approvals	/dashboard/approvals	Approval queue

2. Admin Dashboard

Route: /dashboard/admin



Admin hub — user counts, recent logins, pending approvals, system health.

Tabs

Tab	Purpose
Users	User account management
Roles	Role definitions
Permissions	Permission matrix
Security	Password policy, 2FA, session settings
System Config	Core system parameters
Companies	Multi-company configuration
Approvals	Approval chain configuration
Logs	System audit log

3. User Management

Tab: Users

Route: /dashboard/admin/users

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Admin

Integrations

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Administration
Users, roles, security, system configuration

UsersRolesPermissionsCompaniesSecurityApprovalsCustom FieldsSystem ConfigMobile AppsSystem LogsImport History

Users
8 total

+ Add User

Search name, email, usernanAll Status

NAME	USERNAME	EMAIL	ROLES	TYPE	STATUS	
Chief Executive Officer	ceo	ceo@erp.com	ceo	User	Active	Details →
Marketing Manager	mkt_manager	mktmanager@erp.com	marketing_manager	User	Active	Details →
Chief Marketing Officer	cmo	cmo@erp.com	cmo	User	Active	Details →
System Administrator	admin	admin@erp.local	owner	Superuser	Active	Details →
Chief Technology Officer	cto	cto@erp.com	cto	User	Active	Details →
Chief Financial Officer	cfo	cfo@erp.com	cfo	User	Active	Details →
Data Manager	data_manager	data@erp.com	data_manager	User	Active	Details →
Chief Operating Officer	coo	coo@erp.com	coo	User	Active	Details →

User management — all user accounts with status, role, last login, and actions.

Creating a New User

Click + **New User**:

Field	Required	Notes
Full Name	Yes	
Email	Yes	Used as login username
Password	Yes	Must meet password policy
Role	Yes	Select from configured roles
Department	No	For filtering and approval routing
Employee Link	No	Link to HR employee record
Is Active	Yes	Default true
Must Change Password	No	Force password change on first login

User Status:

Status	Meaning
Active	Can log in
Inactive	Login blocked (retain record)
Locked	Auto-locked after failed logins

Managing Users

- **Reset password** — Admin → Users → select user → Reset Password → sends email link
- **Unlock account** — select user → Unlock
- **Deactivate** — set Is Active = false (does not delete; retains audit history)
- **Never delete** users — deactivate instead. Deletion removes audit trail.

4. Roles & Permissions

Tab: Roles

Route: `/dashboard/admin/roles`

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Administration

Users, roles, security, system configuration

UsersRolesPermissionsCompaniesSecurityApprovalsCustom FieldsSystem ConfigMobile AppsSystem LogsImport History

Roles

35 total

All Status

ROLE	DESCRIPTION	PERMISSIONS	STATUS
owner	Full system access — owner/executive	459 assigned	Active
admin	User and role lifecycle management — plus destructive ERP data operations	103 assigned	Active
ceo	Chief Executive Officer — full cross-functional oversight and import authority	98 assigned	Active
coo	Chief Operating Officer — broad operational oversight	126 assigned	Active
cfo	Chief Financial Officer — full financial and compliance oversight	66 assigned	Active
cto	Chief Technology Officer — integrations, system config, AI, and full import authority	92 assigned	Active
finance_manager	Full finance and M-Pesa access	20 assigned	Active
scoped_hr_manager	HR manager with scoped department and payroll authority	5 assigned	Active
read_only_auditor	Read-only auditor with assigned-scope visibility and optional export	17 assigned	Active

Roles list — system roles with user count and permission summary.

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Administration

Users, roles, security, system configuration

UsersRolesPermissionsCompaniesSecurityApprovalsCustom FieldsSystem ConfigMobile AppsSystem LogsImport History

Permission Matrix

459 permissions · 35 roles

Registry Coverage

Backend registry permissions compared with database permissions.

In sync

Registry

176

Database

459

Covered

176

Missing

0

MISSING REGISTRY PERMISSIONS

None

DATABASE-ONLY PERMISSIONS

ad_performance.edit | ad_performance.view | ai_optimizer.approve | ai_optimizer.run | ai_optimizer.view | analytics.export | analytics.view | attribution.view | audit.export | audit.view | auditor.export | bank_reconciliation.manage_own_scope | bom.ai_all | bom.approve_all | bom.archive_all | bom.cost_all | bom.create_all | bom.delete_all

USERS

7 permissions

Role	Create	Delete	Edit	import	import_template	manage	View
owner	✓	✓	✓	✓	✓	✓	✓
admin	✓	✗	✓	✓	✓	✗	✓
ceo	✗	✗	✗	✓	✓	✗	✗
coo	✗	✗	✗	✓	✓	✗	✗
cfo	✗	✗	✗	✓	✓	✗	✗

Permissions matrix — full list of permissions by module and role.

Role Design

Default roles included:

Role	Typical User
Super Admin	IT Administrator
Finance Manager	Finance lead
Finance User	Accountant / payroll officer
Procurement Manager	Procurement lead
Procurement User	Purchasing officer
Production Manager	Factory/production lead
Production User	Production planner
Quality Manager	Quality lead
Quality User	QC officer
HR Manager	HR lead
HR User	HR officer
Sales Manager	Sales lead
Sales User	Sales rep
Warehouse Manager	Warehouse lead
Warehouse User	Storekeeper
View Only	Auditor / read-only access

Creating a Custom Role

1. Admin → Roles → + **New Role**
2. Set name and description
3. Go to Permissions tab → select role → assign permissions per module
4. Save

Permission granularity: Each module has: view / create / edit / delete / approve / export. Assign only what the role needs.

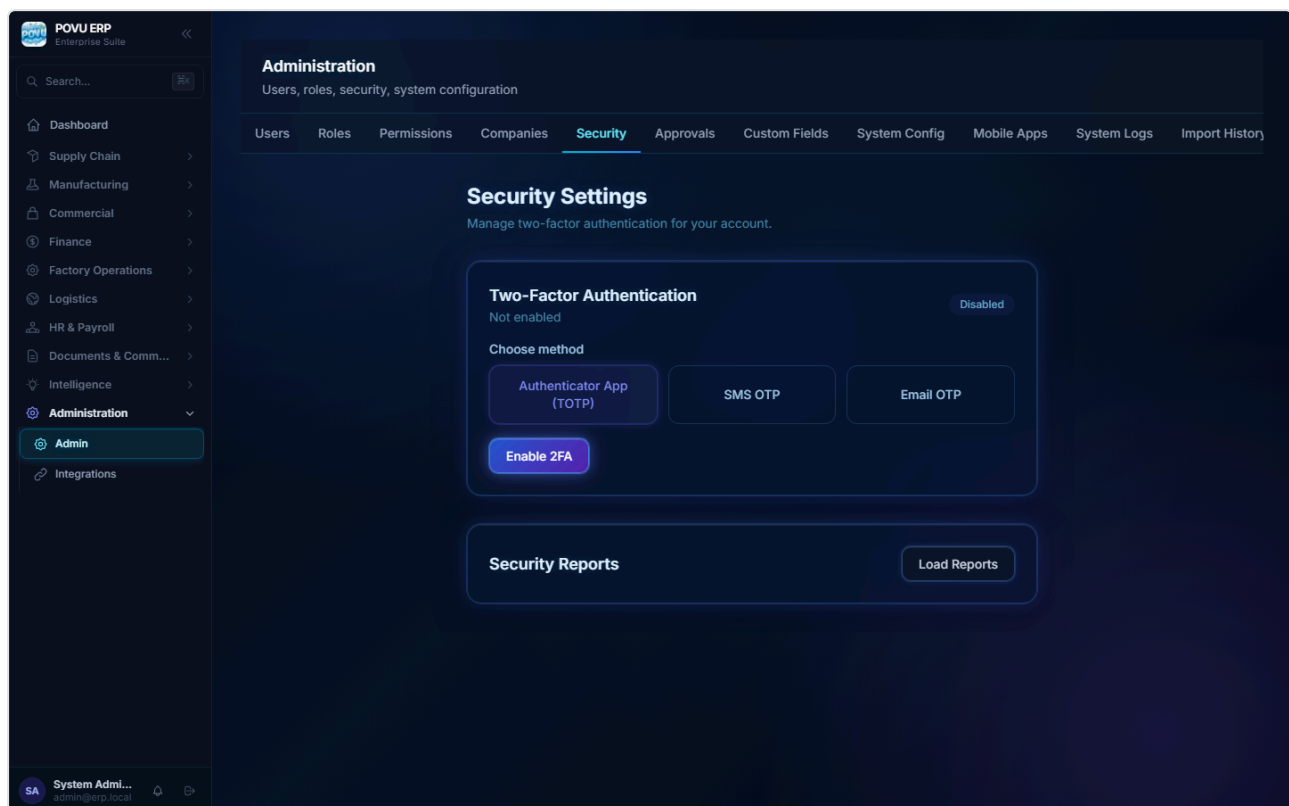
Permission Examples

Permission	Grants
<code>procurement.purchase_orders.approve</code>	Approve POs in Procurement module
<code>finance.journals.post</code>	Post journal entries
<code>hr.payroll.view</code>	View payroll data
<code>inventory.stock.adjust</code>	Post stock adjustments
<code>sales.invoices.void</code>	Void a posted invoice

5. Security Settings

Tab: Security

Route: `/dashboard/admin/security`



Security configuration — password policy, session settings, and 2FA.

Password Policy

Setting	Default	Recommended
Minimum length	8	12
Require uppercase	No	Yes
Require number	No	Yes
Require special char	No	Yes
Expiry (days)	None	90
History (prevent reuse)	None	5

Session Settings

Setting	Default
Session timeout (minutes)	480 (8 hours)
Concurrent sessions	1
Remember me duration	30 days

Failed Login Policy

Setting	Default
Max failed attempts	5
Lockout duration (minutes)	30
Notify admin on lockout	Yes

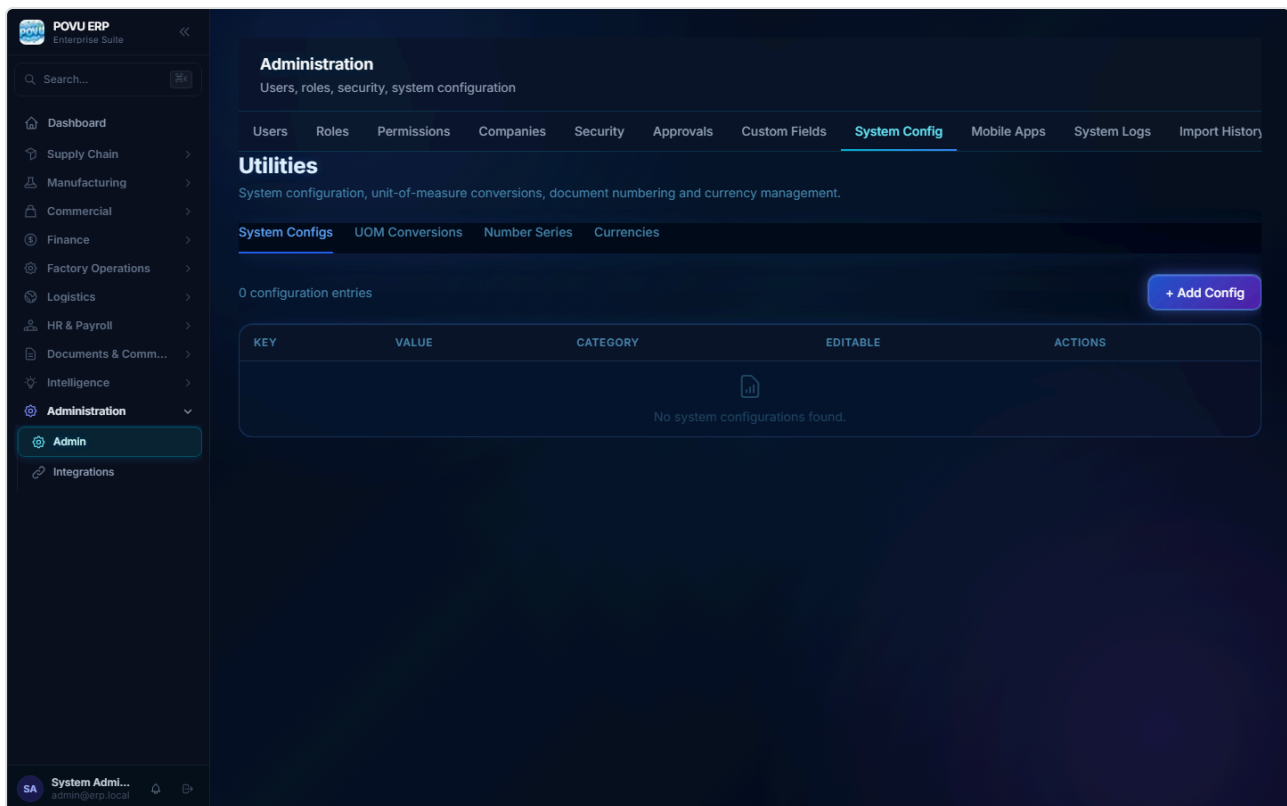
Two-Factor Authentication (2FA)

- Enable 2FA system-wide: Security → 2FA → Enable
- Users enroll via authenticator app (Google Authenticator, Microsoft Authenticator)
- Admin can reset 2FA for a user if they lose their device

6. System Configuration

Tab: System Config

Route: `/dashboard/admin/system-config`



System configuration — timezone, currency, date format, company name, and core settings.

Core Settings

Setting	Notes
Company Name	Displayed on all documents and reports
Default Currency	KES (Kenyan Shilling); can enable multi-currency
Timezone	Africa/Nairobi (UTC+3)
Date Format	DD/MM/YYYY (default for Kenya)
Fiscal Year Start	January (default); change if fiscal year differs
VAT Rate (Standard)	16% (Kenya)
PAYE calculation method	KRA graduated bands
Notification Email	System emails sent from this address
ERP Version	Current version string

Number Series (Document Numbering)

Configure auto-numbering sequences per document type:

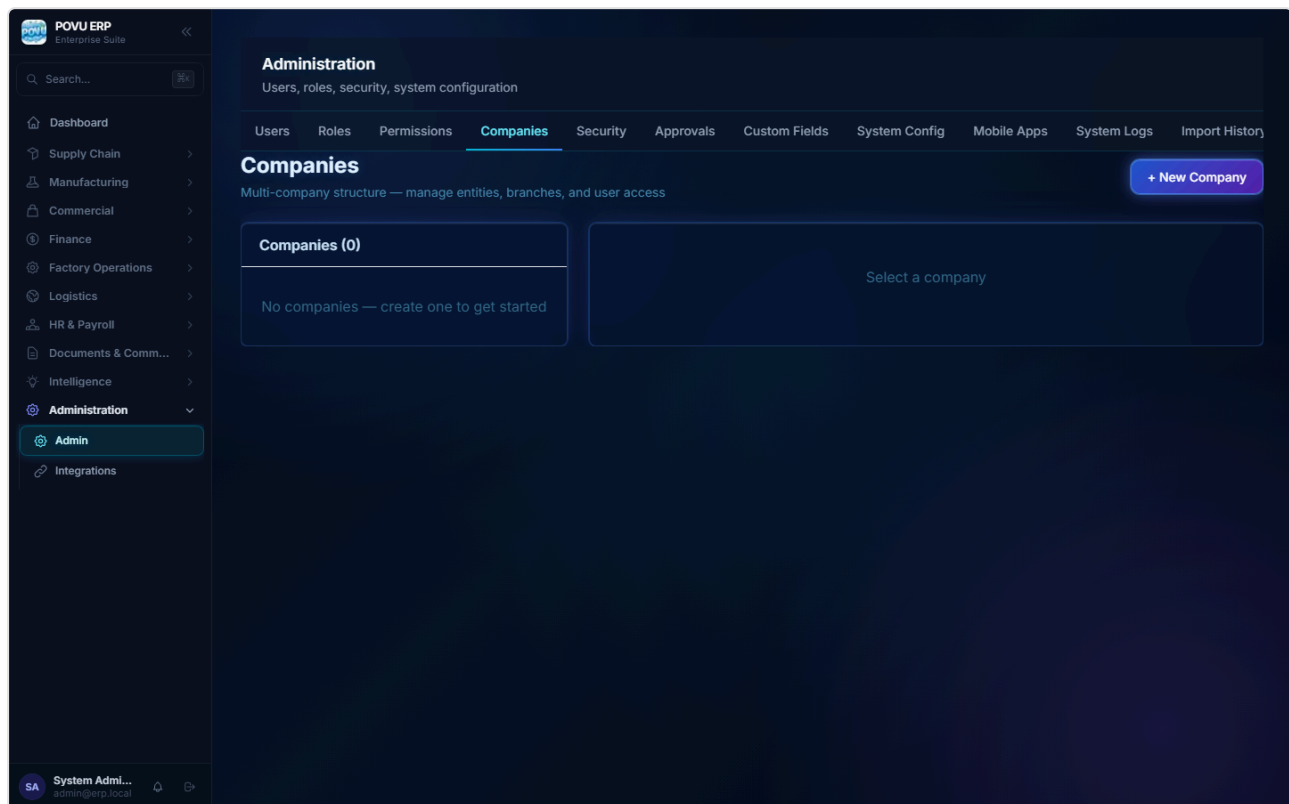
Document	Default Format
Purchase Requisition	PR-{YYYY}-{0001}
Purchase Order	PO-{YYYY}-{0001}
Sales Order	SO-{YYYY}-{0001}
Invoice	INV-{YYYY}-{0001}
Journal	JNL-{YYYY}-{0001}
Work Order	WO-{YYYY}-{0001}

Number series can be reset at year start. Never reuse numbers — breaks audit trail.

7. Company Management

Tab: Companies

Route: `/dashboard/admin/companies`



Company management — multi-entity setup with subsidiary and branch configuration.

Single vs. Multi-Company

- **Single company:** Default setup — one entity, one chart of accounts
- **Multi-company:** Separate legal entities sharing the ERP; intercompany transactions supported

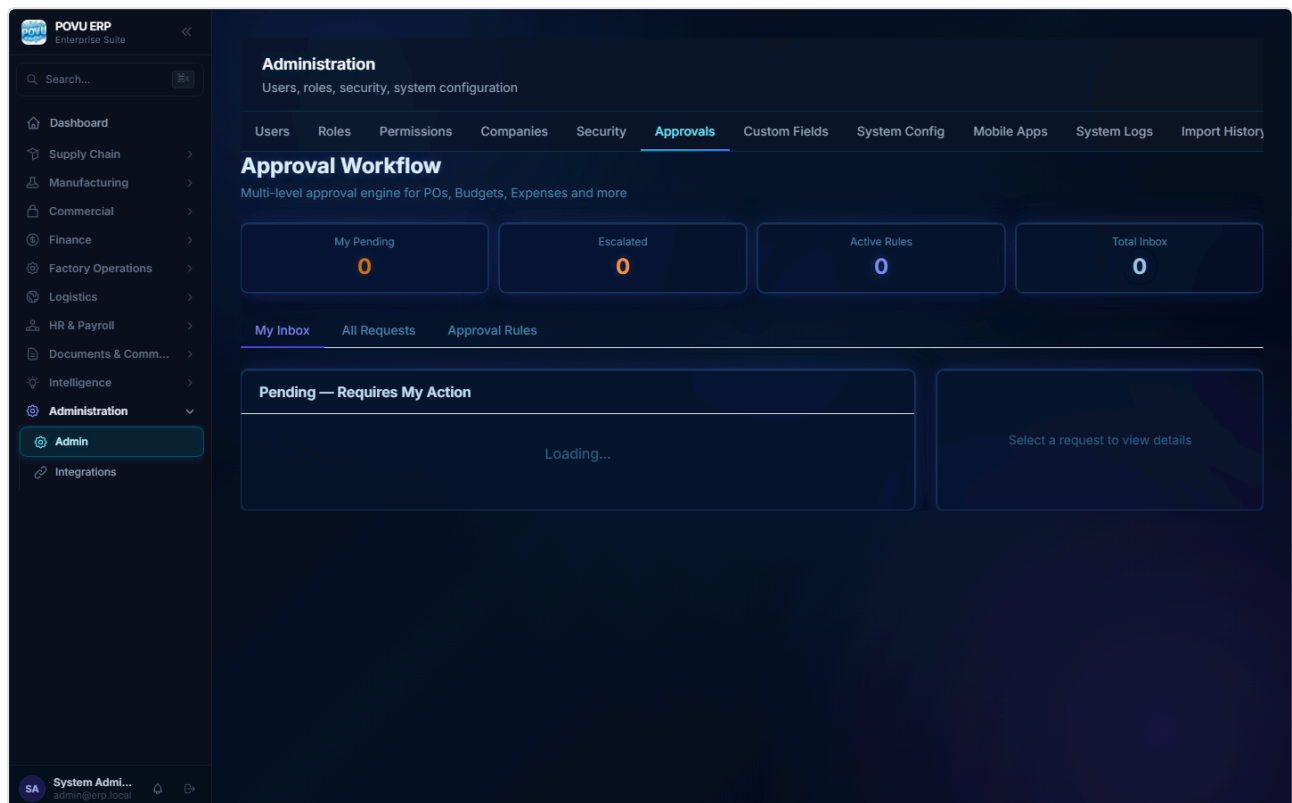
Adding a Company

Field	Required	Notes
Company Name	Yes	Legal entity name
Registration Number	Yes	Companies Registry number
KRA PIN	Yes	Tax identification number
VAT Registration	No	If VAT registered
Address	Yes	Registered address
Currency	Yes	Primary currency
Fiscal Year	Yes	Start month

8. Approval Chain Configuration

Tab: Approvals

Route: `/dashboard/admin/approvals`



Approval chain configuration — define multi-level approvals per document type and amount threshold.

Configuring an Approval Chain

1. Admin → Approvals → + **New Chain**
2. Select document type (PR, PO, Leave, Expense, Journal)
3. Set conditions (e.g. PO Amount > KES 500,000)
4. Add approval levels:
 - Level 1: Approver role or specific user
 - Level 2 (if needed): Escalation approver
 - Parallel approval (if needed): Both approvers must approve

Example PO approval chain:

Condition	Level 1	Level 2
PO < KES 100,000	Procurement Officer	—
PO KES 100,000–500,000	Procurement Manager	—
PO > KES 500,000	Procurement Manager	Finance Director

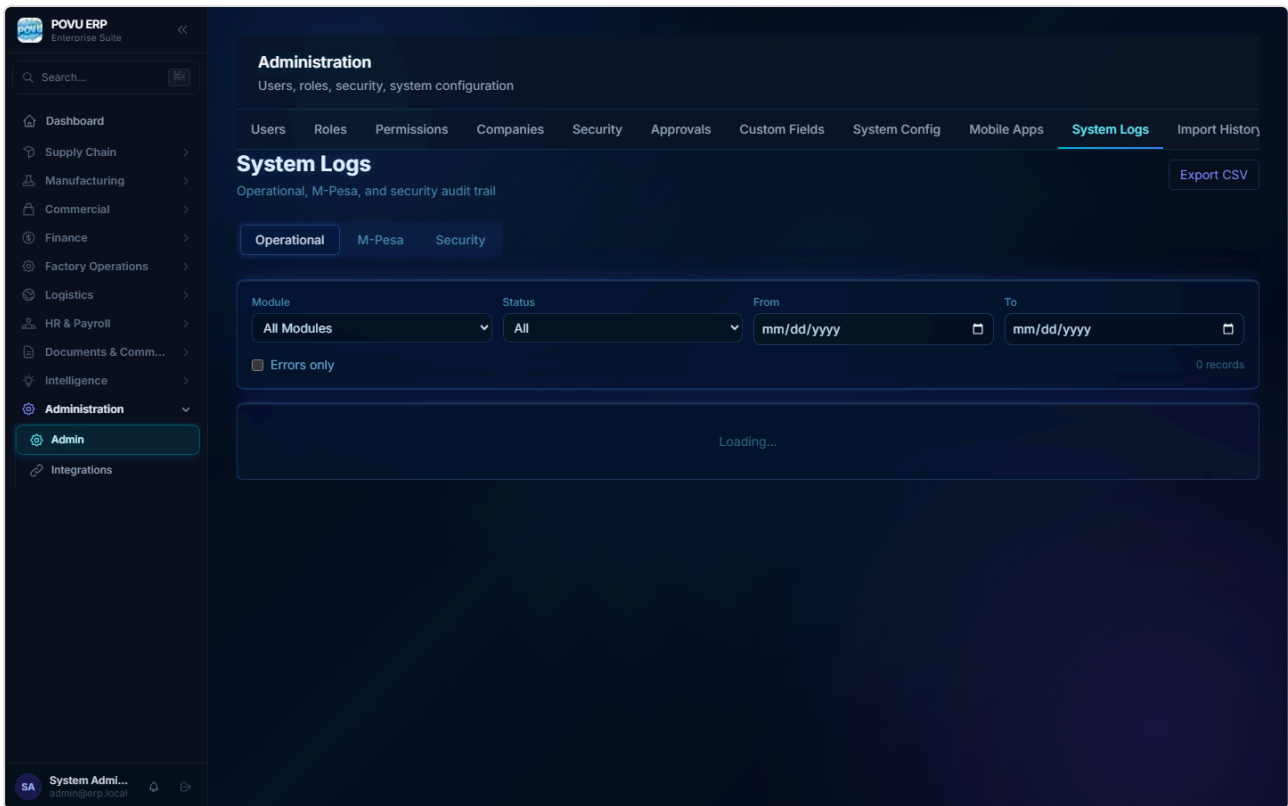
Approval Notifications

Approvers receive email notification when a document enters their queue. Reminder sent after 24 hours if not actioned.

9. Audit Logs (Admin View)

Tab: Logs

Route: `/dashboard/admin/logs`

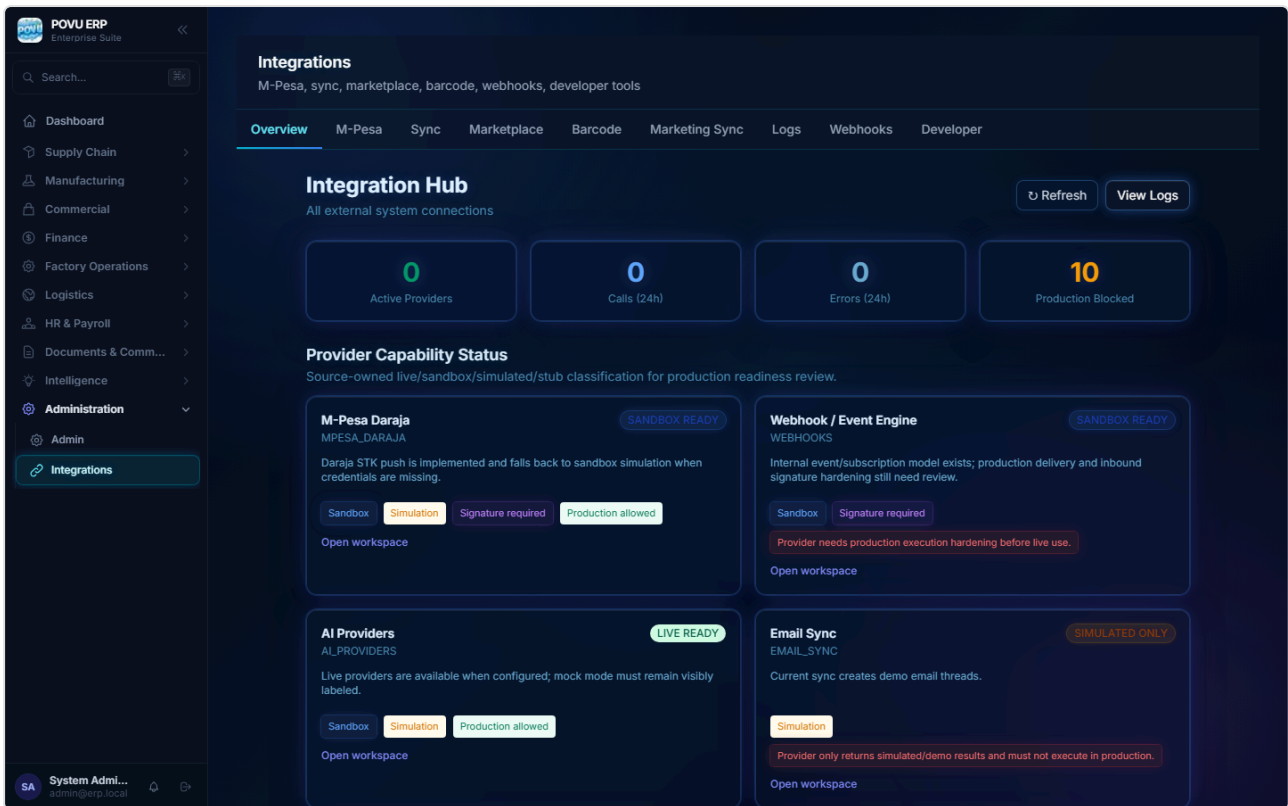


Admin audit log — full system event history with search and filter.

Admin view has full export capability. Standard users can only view logs for their own actions.

10. Integrations

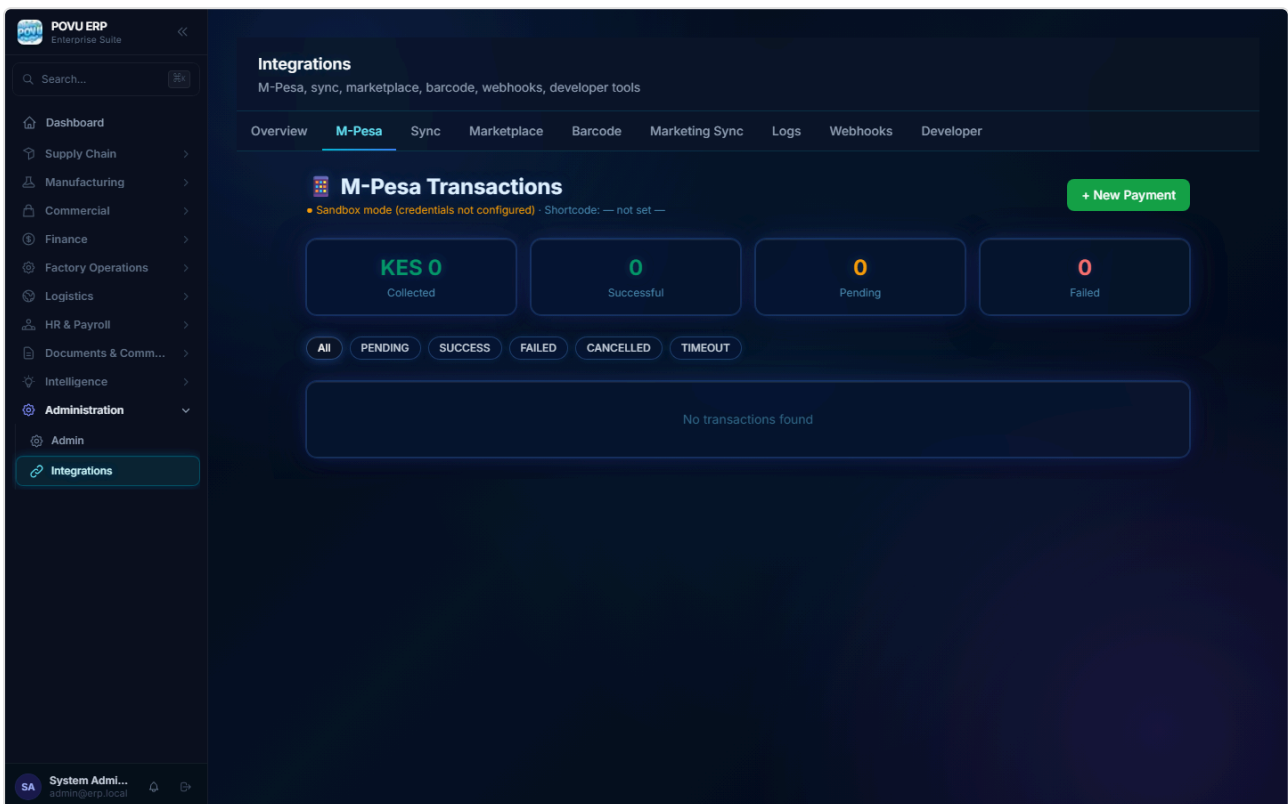
Route: `/dashboard/integrations`



Integrations hub — all external system connections.

M-Pesa Integration

Route: `/dashboard/integrations/mpesa`



M-Pesa Daraja API configuration — consumer key, secret, and shortcode.

Setup steps:

1. Go to Safaricom Developer Portal — create app
2. Get Consumer Key, Consumer Secret, Shortcode
3. Enter in Integrations → M-Pesa
4. Set callback URL to `https://your-erp-domain/api/v1/mpesa/callback`
5. Test with a small transaction
6. Set to **Live** environment

M-Pesa transaction flow:

- Customer pays to shortcode → Daraja callback → ERP auto-receives
- Matched to invoice by reference → invoice marked PAID

eTIMS Integration

Current mode: Simulation (no live KRA calls) **Route:** `/dashboard/finance/etims`

The eTIMS connector is architecturally complete. The SimulationETIMSConnector is active by default — submissions are processed locally with fake ACCEPTED responses.

```
production_execution_allowed = false
```

10 submission statuses: DRAFT, READY, PENDING, SUBMITTED, RETRY_PENDING, ACCEPTED, REJECTED, FAILED, ERROR, CANCELLED

To enable live mode (do not do this without authorization):

1. Select a KRA-approved eTIMS provider
2. Register device at KRA eTIMS developer portal
3. Set `ETIMS_PROVIDER=http`, `ETIMS_API_URL`, `ETIMS_API_KEY`, `ETIMS_PIN`, `ETIMS_BRANCH_CODE` in `.env.production`
4. Obtain accountant approval for GL posting gate rules (TASK-005.1D)

GL posting gate: Finance GL journal posting does NOT currently check eTIMS status — invoices can be posted regardless of REJECTED/ERROR status. This gate is planned but blocked on accountant confirmation.

What admins can do in simulation:

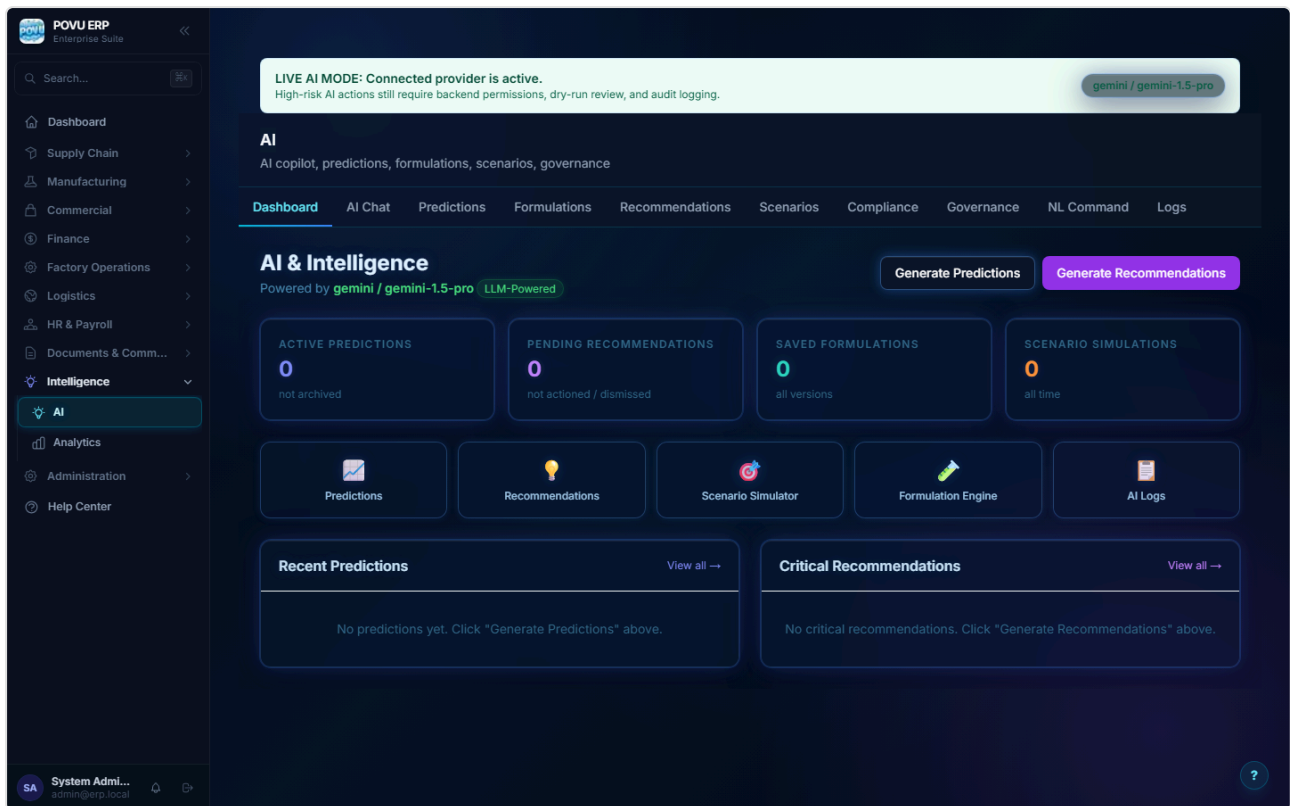
- View all submissions and statuses (Finance → eTIMS tab)
- Retry / Cancel / Poll submissions (routed to simulator)
- Check provider health panel
- View debug payloads per submission

Bank API

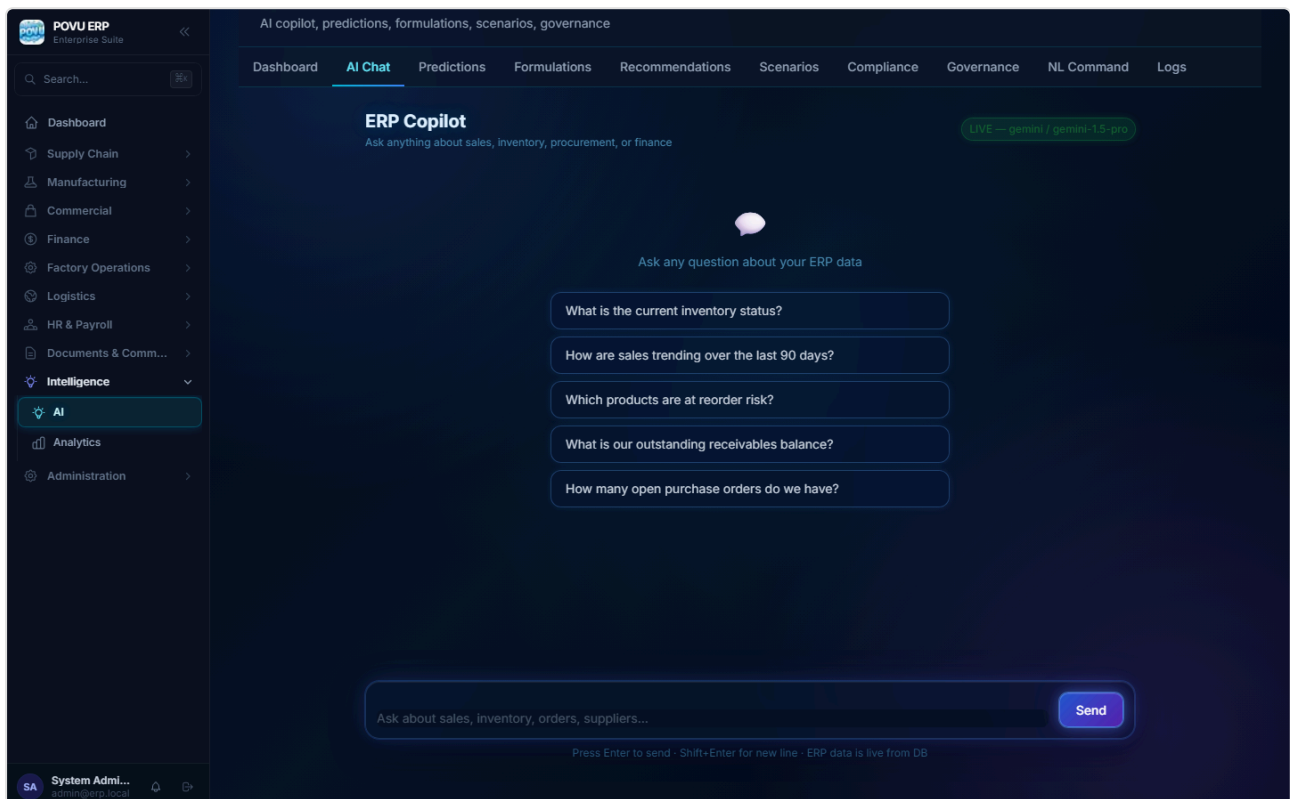
Configured in Finance → Bank API tab. See Finance & Payroll Manual for setup.

11. AI Configuration

Route: `/dashboard/ai`



AI assistant module — intelligent recommendations and automated insights.



AI chat interface — ask questions about your ERP data in natural language.

AI Features

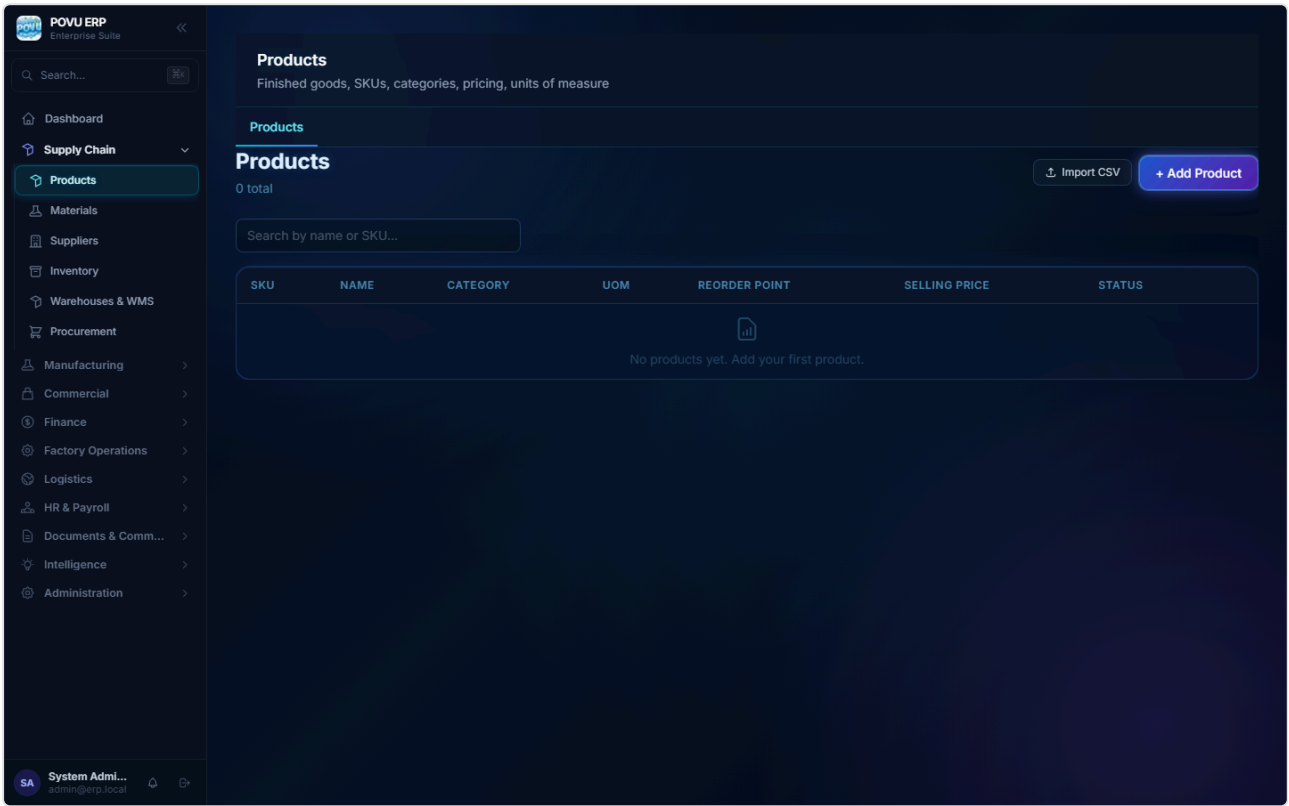
Feature	Description
AI Chat	Ask questions about data: "What is our best selling product this month?"
Demand Forecast	AI-powered sales demand prediction
Procurement Suggestions	Smart reorder recommendations
Quality Alerts	Anomaly detection in QC data
Financial Insights	Automated variance commentary

AI chat examples:

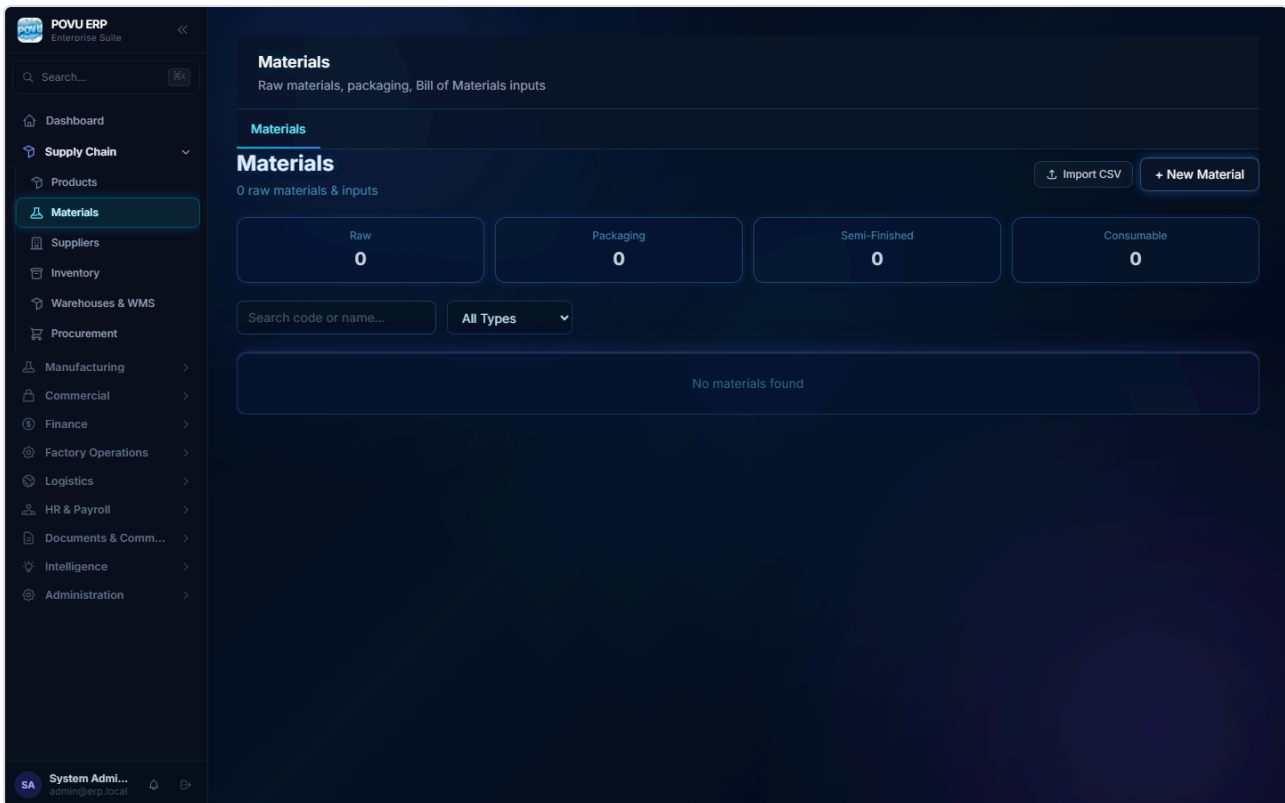
- "Show me all purchase orders above KES 1M this year"
- "Which products are below reorder level right now?"
- "What is the OEE for Line 2 this week?"
- "List employees whose contracts expire in 60 days"

12. Products & Materials Master

Route: /dashboard/products and /dashboard/materials



Products master — finished goods, semi-finished, and trading products.



Materials master — raw materials, packaging, and consumables.

Products

Field	Required	Notes
Product Name	Yes	
SKU	Yes	Unique product code
Product Type	Yes	Finished Good / Semi-Finished / Trading / Service
Category	Yes	Product category
Unit of Measure	Yes	Primary sales UOM
Standard Cost	No	For costing
Sales Price	No	Default sales price
VAT Code	Yes	Tax classification
Is Active	Yes	Default true
Shelf Life (days)	No	For perishable products
Track Batches	No	Enable batch/lot tracking
Min Stock	No	Reorder trigger

Materials

Same fields as Products plus:

- `material_type`: Raw Material / Packaging / Chemical / Consumable
- `purity_spec`: Specification for food-grade materials
- `hazardous`: Boolean flag for dangerous goods

13. Common Mistakes & Troubleshooting

Problem	Cause	Fix
User can't log in	Account locked after failed attempts	Admin → Users → Unlock account
User can't see a module	Missing role permission	Admin → Permissions → add view permission for that module
Approval chain not triggering	Chain not linked to document type	Admin → Approvals → check document type assignment
Document numbering gap	Manual number used	Numbers auto-increment; investigate if gap existed before your change
M-Pesa not receiving	Wrong callback URL	Set callback URL to production URL; not localhost
eTIMS submissions failing	KRA API credentials expired	Renew credentials on KRA eTIMS portal; update in ERP
Two-factor showing for user who lost phone	User unable to enroll	Admin → Users → Reset 2FA → user re-enrolls

14. Security Best Practices

1. **Never share accounts** — each user must have their own login
2. **Principle of least privilege** — grant minimum permissions needed
3. **Enable 2FA** for all admin accounts and finance roles
4. **Audit logs weekly** — review security logs for anomalies
5. **Rotate API keys** — refresh M-Pesa and eTIMS credentials every 6 months
6. **Password policy enforcement** — minimum 12 chars, quarterly rotation for financial roles
7. **Deactivate leavers immediately** — run offboarding checklist the day an employee leaves
8. **Backup regularly** — database backup should run daily; test restores quarterly
9. **Session timeouts** — set to 2 hours for finance roles (shorter than default)

10. **Review user list** monthly — deactivate any account not used in 90 days

15. Security Hardening — Post ERP Updates (v2.1)

Management User Seeding (No Hardcoded Secrets)

Management superuser accounts are created at startup via environment variables:

- `MANAGEMENT_USER_EMAIL`, `MANAGEMENT_USER_USERNAME`, `MANAGEMENT_USER_PASSWORD`, `MANAGEMENT_USER_FULL_NAME`
- If `MANAGEMENT_USER_PASSWORD` is not set, seed exits without creating the account
- **Force password change** is set on first login — initial seed password must be changed before accessing any module

Before go-live: Confirm the management user's initial password has been changed. Check Admin → Users → user record → `force_password_change = false` after first login.

PyJWT Token Security

The system uses PyJWT (not `python-jose`). Do not add `python-jose` to requirements.

Config	Value	Notes
Algorithm	HS256	Symmetric — <code>SECRET_KEY</code> must be secret
<code>SECRET_KEY</code>	Must be 32+ chars random	Generate: <code>openssl rand -hex 32</code>
<code>ACCESS_TOKEN_EXPIRE_MINUTES</code>	Default 30	Increase for less friction; decrease for higher security
Token rejection	Expired tokens return HTTP 401	No grace period

If `SECRET_KEY` is **compromised**: Rotate it immediately. All active sessions are invalidated. Users must log in again.

GS1 Label Generator — Auth-Gated

All 38 GS1 label endpoints require authentication (`require_auth`). Unauthenticated requests return HTTP 401. The frontend GS1 page at `/dashboard/compliance/gs1` requires an active login session.

Do not bypass or remove these guards. GS1 barcodes on product packaging must only be generated by authorized users.

Alembic Migration Advisory Lock

Migrations use `pg_advisory_lock(20260517)` to prevent concurrent migration runs across replicas.

If startup hangs on migrations: Check if another replica holds the lock:

```
SELECT pid, query, state FROM pg_stat_activity WHERE query LIKE '%advisory%';
```

If a dead replica holds the lock, release it:

```
SELECT pg_advisory_unlock(20260517);
```

Then restart the pending replica.

Blocked Live Integrations — Do Not Enable Without Prerequisites

Integration	Status	Required before enabling
eTIMS (KRA)	Simulation	Provider selection + KRA sandbox creds + accountant GL gate approval
M-Pesa (Daraja)	Simulation	Safaricom credentials (Consumer Key, Secret, Shortcode, Passkey, Callback URL)
WhatsApp (Meta)	Demo mode	Meta Cloud API token + Phone Number ID in DB config
AI LLM features	Mock	Anthropic/OpenAI/Gemini API key
SMTP OTP	Dev delivery	Confirm <code>OTP_DEV_DELIVERY_MODE=false</code> in production